QUALITY PREDICTION
IN A MINING PROCESS

Predicting % Silica Concentrate in an Iron Ore Flotation Plant Using Machine Learning

Industrial Internship Report

Bhupesh Singh Mehta | B.Tech CSE – AI/ML

6 Weeks | UniConverge Technologies Pvt Ltd (UCT)

Facilitated by upskill Campus / The IoT Academy

Project Overview

From delayed laboratory quality results to an early ML-based soft sensor

Domain

Mining / Mineral Processing / Industrial AI
Iron ore froth flotation

Dataset

~737,453 observations
24 variables
March–September 2017

Target

% Silica Concentrate
Regression / soft-sensor prediction

Industrial process signals arrive at different sampling rates, while the final silica value comes from delayed laboratory analysis.

The project focuses on predicting silica earlier using available process measurements.

Core challenge: treat the data as a time-series problem and prevent future information from leaking into the model.

Problem Statement & Objectives

Problem

Predict the percentage of silica in the final iron ore concentrate before the delayed laboratory measurement is available. The model should act as a soft sensor that gives an earlier indication of product quality.

Key Objectives

- Explore minute-level prediction feasibility
- Study 1h, 2h, 3h+ forecast horizons
- Compare models with and without % Iron Concentrate
- Identify important process variables
- Balance accuracy, stability, interpretability and computation

Data Preparation & Feature Engineering

Building a leakage-safe time-series modelling table

1. Validate

Schema, types, missing values, duplicates and outliers

2. Parse Time

Convert date/time field and standardize numeric decimal formats

3. Align

Resample and synchronize process signals with the target frequency

4. Engineer

Create lagged and rolling features using only past information

5. Configure

Track A: with % Iron; Track B: without % Iron

Important process groups: feed quality, reagent flows, pulp conditions, flotation air flows and flotation levels.

Proposed ML Pipeline

A modular architecture from raw plant data to monitored predictions

- **Raw Data** — CSV / plant historian
- **Preprocess** — Clean + timestamp
- **Align** — Resample + synchronize
- **Features** — Lags + rolling stats
- **Model** — Regression / XGBoost
- **Output** — Predicted % Silica
- **Monitor** — Error + model health

Models considered: Mean baseline, Linear Regression, Decision Tree, Random Forest and Gradient Boosting/XGBoost.

Evaluation uses chronological train/validation/test separation, with MAE, RMSE, R^2, residuals and feature importance.

Performance Testing & Findings

Evaluation is designed around industrial reliability—not only model fit

Test Areas

- Data integrity
- Baseline performance
- Model comparison
- Minute-level prediction
- Forecast horizons
- No-Iron ablation
- Leakage audit
- Missing/outlier robustness

Metrics

MAE — average absolute error

RMSE — penalizes larger errors

R^2 — variance explained

Also record training time, residual behaviour and feature importance.

Important Result Note

The supplied report does not claim an independently measured final test score because the full 737,453-row CSV was not available for a reproducible executed run.

Published/reference results are used only as benchmarks and guidance.

Learnings & Future Scope

Key Learnings

• Industrial ML is more than algorithm selection
• Time alignment and feature availability are critical
• Leakage prevention is essential in delayed-target problems
• Learned practical Python, Pandas and time-series workflows
• Strengthened model evaluation, research, documentation and presentation skills

Future Work

• Train the full minute-level soft sensor
• Systematic forecast-horizon study
• Compare LSTM/GRU/TCN/Transformer with XGBoost
• SHAP / permutation explainability
• Drift and robustness monitoring
• Streamlit/web dashboard + industrial IoT integration
• Uncertainty estimation and periodic recalibration